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A rank-based approach to improve the efficiency of inferential seamless phase 2/3 clinical trials with dose optimization

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ABSTRACT

To accelerate clinical development, seamless 2/3 adaptive design is an attractive strategy to combine phase 2 dose selection with phase 3 confirmatory objectives. As the regulatory requirement for dose optimization in oncology drugs shifted from maximum tolerated dose to maximum effective dose, it's important to gather more data on multiple candidate doses to inform dose selection. A phase 3 dose may be selected based on phase 2 results and carried forward in phase 3 study. Data obtained from both phases will be combined in the final analysis. In many disease settings biomarker endpoints are utilized for dose selection as they are correlated with the clinical efficacy endpoints. As discussed in Li et al. (2015), the combined analysis may cause type I error inflation due to the correlation and dose selection. Sidák adjustment has been proposed to control the overall type I error by adjusting *p*-values in phase 2 when performing the combined p-value test. However, this adjustment could be overly conservative as it does not consider the underlying correlations among doses/endpoints. We propose an alternative approach utilizing biomarker rank-based ordered test statistics which takes the rank order of the selected dose and the correlation into consideration. If the correlation is unknown, we propose a rank-based Dunnett adjustment, which includes the traditional Dunnett adjustment as a special case. We show that the proposed method controls the overall type I error, and leads to a uniformly higher power than Sidák adjustment and the traditional Dunnett adjustment under all potential correlation scenarios discussed.

1. Introduction

In clinical development of a new therapy, optimal phase 3 dose(s) need to be identified before starting phase 3 confirmatory trials. In non-oncology area, this is usually studied in phase 2 dose-ranging trials which lead to the end of phase 2 discussion with the regulatory agencies. In high unmet medical need areas such as oncology and rare diseases, phase 3 dose(s) are usually identified using phase 1/2 trials combining dose-escalation and dose-expansion phases. Sometimes there may still be insufficient benefit-risk data to fully justify phase 3 dose after these early trials. In particular, FDA has recently recommended dose optimization to justify phase 3 dose in oncology clinical development [2–4]. In particular, the agency requests conducting dose-optimization with PK/PD, efficacy, and safety data across multiple doses to identify the recommended dose for phase 3. To accelerate new drug development in areas of high unmet medical need, inferential seamless phase 2/3 designs may be utilized to combine dose optimization and confirmatory objectives [1,5–7]. In inferential seamless phase 2/3 designs, test statistics from phase 2 and phase 3 are combined for the final primary hypothesis testing to control the overall type I error rate [1,7]. The combined test is usually conducted based on combining inversed normal test-statistics with pre-specified weights and multiplicity-adjustment accounting for multiple dose-selection. When phase 3 efficacy endpoint takes a long follow-up time to mature, biomarkers or early surrogate efficacy endpoints may be utilized for the dose selection in phase 2. Li. et al. (2015) proposed a design to select the dose with the best biomarker response. Sidák adjustment was proposed to control type I error inflation while selecting the best dose at phase 2. Sidák method is a non-parametric *p*-value based approach and doesn't account for the underlying correlation between doses. In practice, optimal dose decision involves review of the totality of data including PK/PD, efficacy, and safety. As a result, the optimal dose is not always the dose with the best response in biomarker or surrogate endpoints. In this case, Sidák adjustment is overly conservative. To address this issue, we have improved the efficiency of adaptive inferential seamless design using rank-based method (instead of Sidák method) for adjusting *p*-value in

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phase 2. The proposed method is a rank-based single-step Dunnett method that controls the overall type I error rate. This method accounts for the rank of the biomarker for selected dose, as well as underlying correlation between doses, therefore it is more powerful compared to Sidák based adjustment.

This manuscript is organized as follows: Section 2 provides a seamless 2/3 design using biomarkers. A biomarker rank-based Dunnett adjustment method is proposed for the overall type I error control that accounts for the underlying correlation between test statistics. Section 3 discusses the underlying joint distribution of test statistics from multiple doses in efficacy endpoints and biomarkers, assuming normal approximation for test statistics. In Section 4, the overall type I error control and power gain using rank-based methods are illustrated via simulations. Section 5 presents an application example in hematologic oncology with progression free survival (PFS) as the primary efficacy endpoint. Section 6 summarizes the proposed method.

2. Rank-based method for overall type I error control in a seamless 2/3 adaptive design

A seamless phase 2/3 clinical trial is a type of adaptive design combining dose-optimization and confirmatory objectives. Multiple doses are often evaluated in Phase 2. An interim analysis (IA) is performed at the end of this phase to select one optimal dose to continue into phase 3, where the efficacy and safety of the selected dose are fully evaluated. The potential advantages of this adaptive design include limiting patient exposure to unsafe or ineffective treatments, saving trial resources, and accelerating clinical development in generating evidence for regulatory decision making [6].

In many disease settings, it's challenging to use efficacy endpoints for phase 2 decision making, due to the long follow-up time required to accumulate these endpoints. In such cases, short-term outcomes such as biomarkers or intermediate endpoints may be utilized to guide dose selection. A phase 2 biomarker endpoint is usually positively correlated with phase 3 efficacy endpoint. The correlation plays an important role in the final hypothesis test that combines p-values of efficacy endpoints in phase 2 and phase 3. Another consideration is the dose selected from phase 2. To balance the overall benefit risk, the selected dose may not always be the dose with the best response in biomarker. Therefore, the rank of biomarker response of the selected dose should also be considered in the multiplicity adjustment for phase 2. With these considerations, we propose a rank-based Dunnett adjustment which accounts for the rank of the biomarker response, as well as correlations between underlying test statistics. The proposed method can be applied to different types of biomarker and efficacy endpoints as long as the test statistic asymptotically follows a normal distribution.

Consider a seamless 2/3 design with 3 doses and 1 control. Stages 1 is phase 2 portion targeting dose optimization. Stage 2 is phase 3 portion for efficacy and safety of the selected dose. Fig. 1 shows the study's overall design.

Based on the totality of phase 2 data, an optimal dose of the investigational drug will be selected. Since the efficacy endpoint (e.g., PFS) is not mature at the end of phase 2, a short-term biomarker is used to facilitate dose selection, together with all observed safety, pharmacokinetics and pharmacodynamics data. In phase 3, efficacy data from both phases will contribute to the confirmatory testing of the efficacy endpoint.

Denote the test statistics of efficacy endpoint and biomarker endpoint as E_j and B_j respectively, where $j = 1, 2, 3$ represents the dose j versus control. The test statistics are assumed to be asymptotically normally distributed. Denote the corresponding ordered biomarker test statistics as $B_{(1)}, B_{(2)}$ and $B_{(3)}$ respectively, where $B_{(1)} < B_{(2)} < B_{(3)}$.

The adjusted p-value for the efficacy testing in subjects in phase 2 will account for the rank r of the biomarker test statistic which is used to select the dose in phase 2. For example, if dose 1 is the selected dose with rank $r = 2$, where $B_{(2)} = B_1$, then the rank based adjusted p-value for the corresponding efficacy test statistic, E_1, is defined as

$$p_1 = Prob(E_1 > e_1),$$

where e_1 is the observed value of E_1. Note that the underlying distribution of E_1 depends on the correlation between biomarker and efficacy endpoints. When the correlation is 0, E_1 follows standard normal distribution; when the correlation is 1, E_1 follows the same distribution as $B_{(2)}$, which is the second order statistics of the 3-dimensional multivariate standard Normal (MVN) distribution.

In general, for a design with m doses, denote the index for the selected dose corresponds to rank r as J_r, where $J_r = 1, 2, \cdots, m$. Then the rank based adjusted p-value for the efficacy of the selected dose in phase 2 is defined as

$$p_1 = Prob(E_{J_r} > e_{J_r}), \tag{1}$$

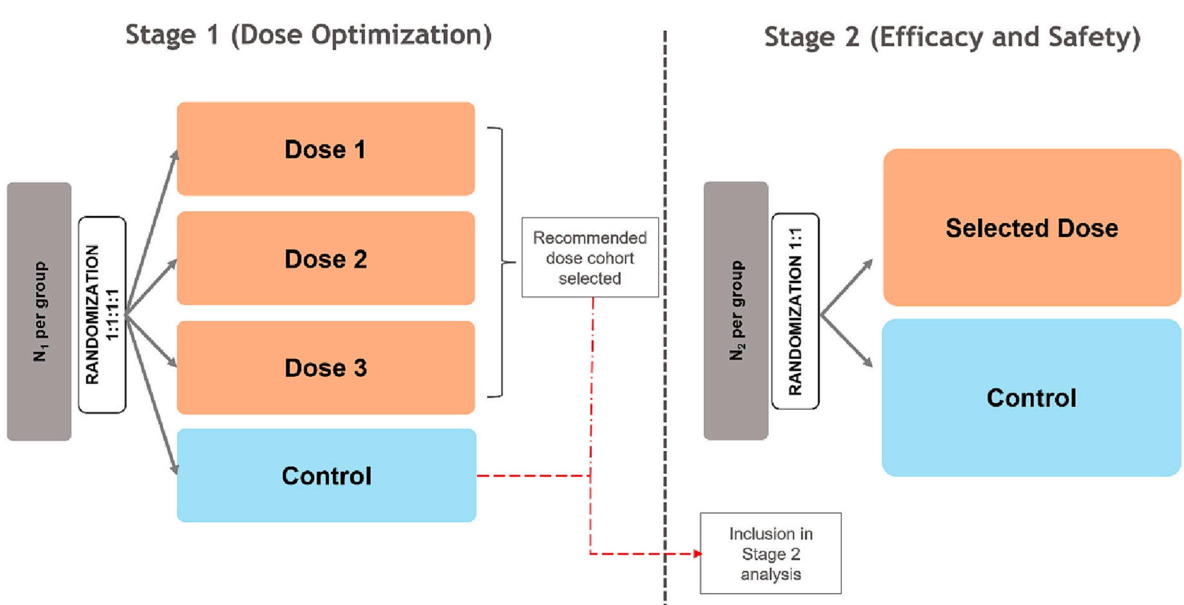

Fig. 1. Overall study design.

where E_{J_r} is the efficacy test statistic corresponding to the selected dose that is associated with $B_{(r)}$ of the m-dimensional MVN distribution, and e_{J_r} is the observed efficacy test statistics of the selected dose. The correlation matrix of the MVN distribution is derived in section 3.2.

We combine the adjusted p-value p_1 and the p-value obtained in phase 3 for the selected dose into a single global test for the final hypothesis test as shown in Formula (2). The test is a weighted inverse normal combination test proposed by Lehmacher and Wassmer (1999). The combined test provides strong control of overall type I error for the final test.

$$p = C(p_1, p_2) = 1 - \Phi\left(\sqrt{w_1}\,\Phi^{-1}(1-p_1) + \sqrt{1-w_1}\,\Phi^{-1}(1-p_2)\right), \tag{2}$$

where p_1 and p_2 are the p-values testing the difference of the efficacy endpoint between the selected treatment group and the control group in phase 2 and 3 respectively; $\Phi(\bullet)$ is the cumulative distribution function (CDF) of normal distribution; $w_1 = \frac{n_1}{n_1+n_2}$, where n_1 and n_2 are the sample sizes of the selected dose group and the control group in phase 2 and phase 3 respectively.

The rank-based Dunnett adjustment has several advantages compared to Sidák adjustment discussed in Li. et al. (2015). Firstly, the biomarker rank of the selected dose is incorporated into calculating the adjusted p-value of phase 2. This leads to dimension reduction when the selected dose is not the one with the best biomarker response. Secondly, the correlation of test statistics among doses is taken into consideration. Thirdly, when the correlation between biomarker and efficacy endpoint is known, it can be incorporated into the calculation of adjusted p-value through the biomarker rank.

In practice, the dose selection at the end of phase 2 is based on the overall benefit-risk assessment. The dose with the best biomarker response may not be the optimal dose for phase 3. It's likely that the selected dose is associated with moderate or smallest biomarker response, based on the totality evaluation of efficacy and safety data. Therefore, rank-based adjustment has the advantage of incorporating this information compared to Sidák adjustment.

To calculate the adjusted p-value using the rank-based method, we evaluated the joint distribution of test statistics across all treatment groups in the efficacy endpoint and the biomarker endpoint. The covariance matrix between the test-statistics of biomarker and efficacy endpoint for m treatment groups versus a control group is derived. This will facilitate simulation of overall type I error rate, as well as simulation of the underlying distribution of the biomarker rank-based test statistics cutoffs for efficacy endpoint in phase 2. Section 3 gives the mathematical setup and results on joint distributions of the test statistics.

3. Joint distribution of test statistics

3.1. Mathematical setup

Denote Y_{ej} and Y_{bj}, $j = 0, 1, 2, \cdots, m$ as the efficacy endpoint and biomarker endpoint for a dose j in phase 2 portion of a seamless 2/3 clinical trial, where $j = 0$ represents the control group, and $j = 1, 2, \cdots, m$ represents the m treatment groups. The sample size of each arm for efficacy endpoint and biomarker endpoint in phase 2 are denoted as n_{ej} and n_{bj} respectively. Let ρ denote the correlation coefficient between Y_{ej} and Y_{bj}. We have

$$E\begin{pmatrix} Y_{ej} \\ Y_{bj} \end{pmatrix} = \begin{pmatrix} \mu_{ej} \\ \mu_{bj} \end{pmatrix},$$

and

$$cov\begin{pmatrix} Y_{ej} \\ Y_{bj} \end{pmatrix} = \begin{pmatrix} \sigma_e^2 & \rho\sigma_e\sigma_b \\ \rho\sigma_e\sigma_b & \sigma_e^2 \end{pmatrix},$$

where σ_e^2 and σ_b^2 are the variances of Y_{ej} and Y_{bj} respectively. To test if there is any difference between Y_{e0} and Y_{ej}, we define the corresponding test statistics as.

$$E_j = \frac{\overline{Y}_{ej} - \overline{Y}_{e0}}{\sqrt{S_e^2\left(\frac{1}{n_{ej}} + \frac{1}{n_{e0}}\right)}},\ j = 1, 2, \cdots, m,$$

where $\overline{Y}_{ej}$ and $\overline{Y}_{e0}$ are the mean of the efficacy endpoints for dose j and control group respectively, S_e^2 is the pooled sample variance.

Similarly, the biomarker test statistics is defined as.

$$B_j = \frac{\overline{Y}_{bj} - \overline{Y}_{b0}}{\sqrt{S_b^2\left(\frac{1}{n_{bj}} + \frac{1}{n_{b0}}\right)}},\ j = 1, 2, \cdots, m,$$

where $\overline{Y}_{bj}$ and $\overline{Y}_{b0}$ are the mean of the biomarker endpoints for dose j and control group respectively, S_b^2 is the pooled sample variance.

We assume the test-statistics follow a standard normal distribution asymptotically under the null hypothesis, i.e., $E_j \sim N(0,1)$ and $B_j \sim N(0,1)$. In practice, the number of subjects used for phase 2 analysis is $n_{ej} = n_{bj} = n_j$, where n_j is the number of subjects enrolled in phase 2.

3.2. Correlation between test statistics

The correlation structure of test statistics E_j and B_j are derived and used to obtain the joint distribution for $j = 1, 2, \ldots, m$. The resulting results are shown below, while the detailed derivation can be found in Appendix 1.

Result 1: The correlation coefficient between efficacy test statistics corresponding to different doses is

$$\rho_e = corr(E_j, E_p) = \sqrt{\frac{n_j n_p}{(n_0 + n_j)(n_0 + n_p)}}, j, p = 1, \cdots, m, j \neq p.$$

The correlation between biomarker test statistics has the same result:

$$\rho_b = corr(B_j, B_p) = \sqrt{\frac{n_j n_p}{(n_0 + n_j)(n_0 + n_p)}}, j, p = 1, \cdots, m, j \neq p.$$

For balanced case, i.e., all groups have the same sample size $n_0 = n_1 = \cdots = n_m = n$, both ρ_e and ρ_b reduces to $\rho_e = \rho_b = \frac{1}{2}$.

Result 2: Given a dose j, the correlation coefficient between test statistics of efficacy endpoint and biomarker endpoint is

$$\rho_{eb1} = Corr(E_j, B_j) = \rho, j = 1, 2, \cdots, m.$$

Note that this correlation coefficient is independent of the sample sizes.

Result 3: The correlation coefficient between test statistics of efficacy endpoint and biomarker given different doses is

$$\rho_{eb2} = Corr(E_j, B_p) = \sqrt{\frac{n_j n_p}{(n_0 + n_j)(n_0 + n_p)}}\rho, j, p = 1, 2, \cdots, m, j \neq p.$$

For balanced case, i.e., all groups have the same sample size, then the correlation coefficient of ρ_{eb2} reduces to $\rho_{eb2} = \frac{1}{2}\rho$.

Consider a balanced design with 3 doses vs. one control, the entire covariance matrix of all test-statistics is illustrated in Table 1.

Under the null hypothesis, $(E_1, E_2, \cdots E_m, B_1, \cdots, B_m)'$ is normally distributed with mean vector 0's and correlation matrix as displayed in Table 1.

In practice, the correlation between efficacy and biomarker endpoints (ρ) is unknown. A shortcut of the proposed method which does not need the information of ρ can be implemented by using the Dunnett adjustment of dimension r. The rank-based Dunnett adjusted p-value for the selected dose (J_r) is defined as

$$p_1 = Prob\left(T_{(r)} > e_{J_r}\right), \tag{3}$$

Table 1
The correlation matrix of test statistics (TS).

Test Statistics		Efficacy TS			Biomarker TS		
		E_1	E_2	E_3	B_1	B_2	B_3
Efficacy TS	E_1	1	$\frac{1}{2}$	$\frac{1}{2}$	ρ	$\frac{1}{2}\rho$	$\frac{1}{2}\rho$
	E_2	$\frac{1}{2}$	1	$\frac{1}{2}$	$\frac{1}{2}\rho$	ρ	$\frac{1}{2}\rho$
	E_3	$\frac{1}{2}$	$\frac{1}{2}$	1	$\frac{1}{2}\rho$	$\frac{1}{2}\rho$	ρ
Biomarker TS	B_1	ρ	$\frac{1}{2}\rho$	$\frac{1}{2}\rho$	1	$\frac{1}{2}$	$\frac{1}{2}$
	B_2	$\frac{1}{2}\rho$	ρ	$\frac{1}{2}\rho$	$\frac{1}{2}$	1	$\frac{1}{2}$
	B_3	$\frac{1}{2}\rho$	$\frac{1}{2}\rho$	ρ	$\frac{1}{2}$	$\frac{1}{2}$	1

where e_{J_r} is the observed efficacy test statistic of the selected dose, $T_{(r)}$ is the rth order statistic of the m-dimensional t distribution. Under normal approximation, this is equivalent to setting degree of freedom to be infinity. Note that $T_{(r)}$ is different from E_{J_r} in formula (1), where E_{J_r} is the efficacy test statistic for the selected dose, with its distribution depending on 2m-dimensional standard MVN distribution with known correlation matrix (e.g. Table 1). This rank-based Dunnett adjusted p-value is generally larger than the rank-based adjusted p-value calculated under known correlation.

4. Simulations for type I error control and power

In this section we present simulation studies comparing the proposed rank-based Dunnett method and the Sidák adjustment as proposed in Li et al. (2015). The correlation matrix is used to simulate efficacy and biomarker test statistics in phase 2 sampling from the joint distribution. The function *mvrnorm()* in the R - MASS package is used for this purpose: *z1 = mvrnorm(n = B, mu = rep(0, 2*j), Sigma = sig)*, where B is the number of simulations, *j* is the number of treatment groups, and *sig* is the correlation matrix, as shown in Table 1. Under the null hypothesis, *mu* is set to 0 for all test statistics. A full range of possible correlation coefficients ρ from 0 to 1 is evaluated in the simulation, by incremental step of 0.1. The code *z2 = rnorm(n = B)* is used to generate the simulated efficacy test statistics in phase 3.

To adjust the p-value p_1, we use the rank-based Dunnett adjustment which relies on the rank r of biomarker test statistic. The function used for this adjustment is *AdjustPvalues(pval, proc = "DunnettAdj", par = parameters(n = Inf))* in the *Mediana* R-package. The *pval* in the argument is set up as $pval = c(p_{01}, rep(1, r-1))$, where p_{01} is the p-value before adjustment. If the selected dose has the smallest biomarker test statistics, i.e., $r = 1$, no adjustment is conducted for the original p-value. However, if the selected dose group has a rank $1 < r < m$,then a r-group single-step Dunnett adjustment will be applied to the p-value accordingly. If $r = m$, the number of doses, the rank-based Dunnett adjustment is identical to the traditional Dunnett adjustment. We used $pval = c(p_{01}, rep(1, m-1))$ to simulate the traditional Dunnett adjusted p values given different r.

Let the sample sizes be 50 and 300 per group in phase 2 and 3 respectively. Then the pre-specific weight is $w_1 = \frac{50}{50+300} = \frac{1}{7}$. We use $m = 3$, $B = 50,000$ as an example to demonstrate the simulation results for the overall type I error rate and power.

4.1. Overall type I error rate simulation results

The family-wise error rate (FWER) based on Sidák and Rank-based Dunnett adjustments are shown in the Table 2. The results show that the rank-based Dunnett adjustment can control the type I error at 0.025 across all different $\rho' s$ and $r' s$. In practice, ρ is usually unknown, thus the actual type I error can vary from 0.0126 to 0.025.

The results also show that rank-based Dunnett adjustment is always less conservative than the Sidák adjustment and traditional Dunnett adjustment for $r < 3$. This is as expected since the proposed method accounts for the correlation between different doses/endpoints, as well as reduced dimension of multiplicity adjustment based on observed biomarker rank r. When $r = 3$, the rank-based Dunnett adjustment is identical to traditional Dunnett adjustment.

4.2. Empirical cutoffs of rank based test statistics

We noticed that across all scenarios in Table 2, the simulated overall type I error rates are less than the nominal level of 0.025. To further examine the overall type I error control performance of the rank-based Dunnett method, empirical cumulative distribution functions (ECDF) of the true ordered efficacy test statistics E_{J_r} and Dunnett adjusted efficacy test statistics were simulated, respectively. Empirical critical values for $\alpha = 0.025$ are listed in Table 3. The number of runs for a simulation is 50,000. The three columns under "ρ is known" are the simulated observed critical values using rank-based exact adjustment method for a given ρ. The cutoffs under "ρ is unknown," are for rank-based Dunnett adjustment and Sidák adjustment respectively. The cutoff for Sidák adjustment is 2.3909 showing at the row of $r = m = 3$ which is the largest value in the table. The other two values 1.9600 and 2.2390 correspond to the scenarios by letting $m = 1$ and $m = 2$. The cutoffs for the short-cut Dunnett method are 1.9603, 2.2132, and 2.3487 for $r = 1$, 2, and 3 respectively. They are same across all correlation $\rho' s$ given a r.They are larger than those of the exact α-level cutoffs in most cases. The directional trend is consistent with those simulated FWERs in Table 2. For example, when $r = 2$, the rank-based Dunnett method is associated with larger cutoffs across all values of ρ. When $\rho = 0$, the α-level cutoff is the same as N(0,1); when $\rho = 1$, the α-level cutoff is the same as the r-th ordered statistics (e.g., r=3, E_{J_r} cutoff = Dunnett cutoff).

Table 2
FWER on test statistic-level simulation results.

ρ	Sidák Adjustment			Traditional Dunnett Adjustment			Rank-based Dunnett Adjustment		
	$r = 1$	$r = 2$	$r = 3$	$r = 1$	$r = 2$	$r = 3$	$r = 1$	$r = 2$	$r = 3$
0	0.0103	0.0103	0.0107	0.0135	0.0138	0.0135	0.0230	0.0172	0.0135
0.1	0.0109	0.0121	0.0126	0.0141	0.0149	0.0159	0.0243	0.0179	0.0159
0.2	0.0099	0.0122	0.0136	0.0127	0.0153	0.0170	0.0215	0.0181	0.0170
0.3	0.0095	0.0122	0.0147	0.0129	0.0157	0.0185	0.0218	0.0187	0.0185
0.4	0.0082	0.0117	0.0149	0.0114	0.0152	0.0186	0.0197	0.0179	0.0186
0.5	0.0078	0.0112	0.0165	0.0106	0.0149	0.0200	0.0188	0.0178	0.0200
0.6	0.0067	0.0107	0.0163	0.0093	0.0141	0.0201	0.0168	0.0169	0.0201
0.7	0.0058	0.0100	0.0175	0.0083	0.0138	0.0215	0.0161	0.0170	0.0215
0.8	0.0054	0.0096	0.0185	0.0076	0.0130	0.0217	0.0145	0.0159	0.0217
0.9	0.0047	0.0097	0.0193	0.0074	0.0127	0.0230	0.0137	0.0155	0.0230
1	0.0040	0.0093	0.0200	0.0062	0.0125	0.0242	0.0126	0.0153	0.0242

Footnote: When $r = 3$, the rank-based Dunnett adjustment is identical to the traditional Dunnett adjustment.

Table 3
Comparison of critical values (one-sided $\alpha = 0.025$).

Correlation Biomarker & Efficacy	ρ is known			ρ is unknown		
	E_{J_r} Cutoff			Rank-based Dunnett Cutoff // Sidák Cutoff		
	$r=1$	$r=2$	$r=3$	$r=1$	$r=2$	$r=3$
$\rho=0$	1.9555	1.9626	1.9629	1.9603	2.2132	2.3487
$\rho=0.1$	1.8961	1.9558	2.0144	//	//	//
$\rho=0.2$	1.8322	1.9480	2.0712	1.9600	2.2390	2.3909
$\rho=0.3$	1.7613	1.9373	2.1262			
$\rho=0.4$	1.6922	1.9166	2.1661			
$\rho=0.5$	1.5986	1.8858	2.2093			
$\rho=0.6$	1.5146	1.8575	2.2416			
$\rho=0.7$	1.4295	1.8228	2.2754			
$\rho=0.8$	1.3325	1.7800	2.3044			
$\rho=0.9$	1.2262	1.7275	2.3323			
$\rho=1$	1.1152	1.6699	2.3473			

When $0 < \rho < 1$, the cutoffs are bounded between those of 0 and 1.

Table 3 also shows that the rank-based method cut-off values goes up as ρ increases for $r = 3$, however the cut-off values go down as ρ increases for $r = 1, 2$. This is because when $r = 3$, the dose with the maximum biomarker response is selected. As ρ increases, this means the dose is also more likely to have maximum efficacy response. Therefore, the cutoffs for E_{J_r} should be adjusted up to account for the 'cherry-picking' the best outcome.

On the other hand, when $r = 1$, the dose with minimum biomarker response is selected. As ρ increases, this means the dose is also more likely to have minimum efficacy response. Therefore, the cutoffs for E_{J_r} should be adjusted down to account for selecting the worse outcome. When $r = 2$, this is a middle scenario between $r = 1$ and $r = 3$. As a result, the change in cutoffs is less in magnitude as compared to the other 2 extreme cases. The downward trend could reflect the higher chance of selecting the efficacy outcome towards the minimum end.

4.3. Power simulation

We use an efficacy endpoint of PFS as an example to demonstrate the power of the rank-based Dunnett adjustment. The log-rank test statistic is approximately normal, $N\left(-log(\mathrm{HR})\sqrt{E/4}, 1\right)$, where HR is the hazard ratio, and E is the expected number of events at the time of the analysis. Assume HR = 0.7 under the alternative hypothesis. To obtain a 90% or higher power, at least 331 events are required in a simple two-sample test design without interim analyses (EAST 6.2). Assume a binary biomarker endpoint is obtained at the end of phase 2, where the expected response rates (RR) for the control group is $RR_c = 0.5$, and for treatment groups are $RR_{t1} = 0.6$, $RR_{t2} = 0.65$, and $RR_{t3} = 0.7$ respectively. The test statistic for the biomarker endpoint RR follows a normal distribution with mean $\frac{RR_{tj}-RR_c}{\sqrt{\left[RR_{tj}\cdot(1-RR_{tj})+RR_c\cdot(1-RR_c)\right]/n_1}}$ for $j = 1, 2, 3$, and correlation coefficient $\frac{1}{2}$ as discussed in section 3.2. Table 4 shows the simulated power of the rank-based Dunnett adjustment falls between 86% and 90%, which is uniformly more powerful than Sidák adjustment and traditional Dunnett adjustment for $r < 3$. When $r = 3$, the rank-based Dunnett adjustment is identical to traditional Dunnett adjustment, which is more powerful than Sidák adjustment.

5. Application

We apply the rank-based Dunnett method to a clinical study example where an inferential seamless phase 2/3 adaptive design as described in Section 2 is adopted.

Instead of simulation based on test statistics as shown in section 4, a subject-level simulation is performed to assess the operational characteristic of the rank-based Dunnett method. PFS is used for both dose selection and confirmatory efficacy testing, which corresponds to $\rho = 1$ in Table 2.

Assuming the PFS is exponentially distributed with a median of 20 months in the control group and HR = 0.7 under the alternative hypothesis, 331 events are required to obtain 90% power in a simple two-sample test design (EAST 6.2).

In simulation, n_1=50 subjects are simulated in four arms separately in phase 2. In phase 3, $n_2 = 300$ subjects are simulated in the selected dose group and control group separately with a constant enrollment rate of 40 per month. The first interim analysis for dose selection will occur 3 months after the 200th subject is enrolled. The PFS log-rank tests are conducted between the three doses and control, separately. One dose group with the smallest numerical p-value will be selected to continue into phase 3. In phase 3, one superiority interim analysis at 75% PFS information (and a final analysis at 100% PFS information are conducted. O'Brien-Fleming α-Spending function is used to determine the nominal p-value in the analysis. Table 5 shows the type I error and power simulation results based on 50,000 runs. Three rows correspond to three scenarios of dose group selection based on their observed PFS ranks at IA1. The type I error is well controlled under 0.025 across all scenarios for the proposed method. The power is around 87–88% for the rank-based Dunnett adjustment, compared to 79–86% for the Sidák adjustment, and 82%–88% for the traditional Dunnett adjustment. The power gain of the proposed method is more pronounced when rank $r = 1, 2$, where it accounts for both Phase 2 endpoint rank and the between-dose correlation. When rank $r = 3$, the power gain over the Sidak adjustment is the same as applying the traditional Dunnett adjustment since the rank-based Dunnett adjustment is identical to traditional Dunnett adjustment.

In addition, we compare the inferential seamless design with conducting separate Phase 2 and Phase 3 trials, where Phase 2 data is used

Table 4
Test Statistic-level simulation results for power.

ρ	Sidák Adjustment			Traditional Dunnett Adjustment			Rank-based Dunnett Adjustment		
	$r=1$	$r=2$	$r=3$	$r=1$	$r=2$	$r=3$	$r=1$	$r=2$	$r=3$
0	0.8281	0.8288	0.8273	0.8557	0.8559	0.8549	0.8990	0.8736	0.8549
0.1	0.8251	0.8313	0.8360	0.8536	0.8585	0.8611	0.8980	0.8755	0.8611
0.2	0.8185	0.8282	0.8417	0.8497	0.8554	0.8662	0.8954	0.8726	0.8662
0.3	0.8146	0.8313	0.8487	0.8452	0.8582	0.8714	0.8917	0.8745	0.8714
0.4	0.8062	0.8301	0.8496	0.8385	0.8563	0.8730	0.8870	0.8725	0.8730
0.5	0.8003	0.8308	0.8565	0.8340	0.8576	0.8764	0.8832	0.8739	0.8764
0.6	0.7941	0.8321	0.8626	0.8299	0.8589	0.8828	0.8817	0.8753	0.8828
0.7	0.7921	0.8347	0.8715	0.8285	0.8611	0.8882	0.8811	0.8777	0.8882
0.8	0.7834	0.8344	0.8734	0.8215	0.8590	0.8902	0.8753	0.8757	0.8902
0.9	0.7769	0.8325	0.8771	0.8164	0.8589	0.8927	0.8714	0.8751	0.8927
1	0.7702	0.8361	0.8829	0.8106	0.8613	0.8974	0.8683	0.8780	0.8974

Footnote: When $r = 3$, the rank-based Dunnett adjustment is identical to the traditional Dunnett adjustment.

Table 5
Subject-level simulation results for PFS.

Rank	Type I Error			Power			
	Sidák Adjustment	Traditional Dunnett Adjustment	Rank-based Dunnett Adjustment	Sidák Adjustment	Traditional Dunnett Adjustment	Rank-based Dunnett Adjustment	Phase 3 only
$r = 1$	0.0072	0.0096	0.0179	0.7865	0.8191	0.8702	0.8226
$r = 2$	0.0114	0.0144	0.0175	0.8274	0.8510	0.8674	0.8237
$r = 3$	0.0160	0.0198	0.0198	0.8625	0.8804	0.8804	0.8264

Footnote: When $r = 3$, the traditional Dunnett adjustment is identical to Rank-based Dunnett adjustment.

for dose selection and Phase 3 data is used for confirmatory efficacy and safety analysis. The 'Phase 3 Only' column in Table 5 shows the study power based on only Phase 3 subjects are 4–5% lower than the rank-based Dunnett adjustment based on the inferential seamless design combining both phase 2 and 3 subjects. To maintain the same power as the rank-based Dunnett adjustment, approximately 60 to 75 more subjects will need to be enrolled in Phase 3, assuming an event rate of 60%.

6. Summary and discussion

Inferential seamless phase 2/3 designs are useful to accelerate drug development by combining phase 2 dose optimization objective with phase 3 confirmatory objective. Proper multiplicity adjustment should be applied to control the overall type I error. Depending on data availability at the end of phase 2, efficacy endpoint or surrogate biomarker response may be used for dose selection. As the recommended dose may not be the dose with the best efficacy response at the end of phase 2 due to overall benefit-risk consideration, we propose an alternative multiplicity adjustment method based on the rank of phase 2 responses. The rank-based method is applicable to dose selection based on an efficacy endpoint or a biomarker positively correlated with the efficacy endpoint.

Under normal approximation, the correlation matrix of test statistics for all doses in efficacy endpoint and biomarker is derived. This allows the simulation of joint distribution of all test statistics at the end of phase 2 under the null and alternative hypothesis. When the correlation between the efficacy endpoint and biomarker is known, one can obtain the corresponding cut-off for the α-level test numerically by simulating the empirical cumulative distribution function of the rank-based test statistics.

The rank-based Dunnett methods can be applied to the scenarios when correlation between endpoints is unknown. Since this method accounts for both the rank of phase 2 biomarker responses and the correlation between test statistics among doses, it is uniformly more powerful than the Sidák adjustment and traditional Dunnett adjustment.

Simulation results suggested that the proposed method is still conservative. For example, when the second ranked dose is selected, the simulated overall type I error rate was always less than the target level of 0.025, regardless of the correlation between the efficacy endpoint and the biomarker. When the minimal ranked dose is selected, the test becomes more conservative as the correlation between efficacy endpoint and biomarker increases, and vice versa for the maximum ranked dose. This implies that the knowledge of the correlation between the efficacy and the biomarker endpoints is an important factor in determining the exact α-level cutoff. Further research may be conducted on incorporating the knowledge of efficacy-biomarker correlation into the test.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

We don't use any real data. Simulation code will be available on request.

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Appendix A. Proof of correlation between test statistics

Formulas that we use for the coefficient calculations are listed below.

$$Cov(X, Y) = E(XY) - E(X)E(Y),$$

$$Cov(\overline{X}, \overline{Y}) = \frac{1}{n_1 n_2}\sum_{i=1}^{n_1}\sum_{j=1}^{n_2} Cov(X_i, Y_j),$$

$$E(X^2) = Var(X) + E(X)^2,$$

$$Var(X \pm Y) = Var(X) + Var(Y) \pm 2Cov(X, Y).$$

Result 1: The correlation coefficient between efficacy test statistics corresponding to different doses

$$\rho_e = Corr(E_j, E_p)$$
$$= Corr\left(\overline{Y}_{ej} - \overline{Y}_{e0},\ \overline{Y}_{ep} - \overline{Y}_{e0}\right)$$
$$= \frac{Cov\left(\overline{Y}_{ej} - \overline{Y}_{e0},\ \overline{Y}_{ep} - \overline{Y}_{e0}\right)}{\sqrt{Var\left(\overline{Y}_{ej} - \overline{Y}_{e0}\right)\cdot Var\left(\overline{Y}_{ep} - \overline{Y}_{e0}\right)}}$$
$$= \frac{Var\left(\overline{Y}_{e0}\right)}{\sqrt{Var\left(\overline{Y}_{ej}\right) + Var\left(\overline{Y}_{e0}\right)}\cdot\sqrt{Var\left(\overline{Y}_{ep}\right) + Var\left(\overline{Y}_{e0}\right)}}$$
$$= \frac{\frac{1}{n_{e0}}\cdot\sigma_e^2}{\sigma_1^2\sqrt{\frac{1}{n_{ej}} + \frac{1}{n_{e0}}}\cdot\sqrt{\frac{1}{n_{ep}} + \frac{1}{n_{e0}}}} = \sqrt{\frac{n_j n_p}{(n_0 + n_j)(n_0 + n_p)}},$$

for $j, p = 1, \cdots, m, j \neq p$.

The correlation between biomarker test statistics has the same result:

$$\rho_b = corr(B_j, B_p) = \sqrt{\frac{n_j n_p}{(n_0 + n_j)(n_0 + n_p)}}, j, p = 1, \cdots, m, j \neq p.$$

For balanced case, i.e., all groups have the same sample size $n_0 = n_1 = \cdots = n_m = n$, both ρ_e and ρ_b reduces to $\rho_e = \rho_b = \frac{1}{2}$.

Result 2: Given a dose j, where $j = 1, 2, \cdots, m$, the correlation coefficient between test statistics of efficacy endpoint and biomarker endpoint is

$$\rho_{eb1} = Corr(E_j, B_j) = Corr\left(\overline{Y}_{ej} - \overline{Y}_{e0},\ \overline{Y}_{bj} - \overline{Y}_{b0}\right)$$
$$= \frac{Cov\left(\overline{Y}_{ej} - \overline{Y}_{e0},\ \overline{Y}_{bj} - \overline{Y}_{b0}\right)}{\sqrt{Var\left(\overline{Y}_{ej} - \overline{Y}_{e0}\right)\cdot Var\left(\overline{Y}_{bj} - \overline{Y}_{b0}\right)}}$$
$$= \frac{Cov\left(\overline{Y}_{ej}, \overline{Y}_{ej}\right) + Cov\left(\overline{Y}_{b0}, \overline{Y}_{b0}\right)}{\sqrt{\left[Var\left(\overline{Y}_{ej}\right) + Var\left(\overline{Y}_{e0}\right)\right]\cdot\left[Var\left(\overline{Y}_{bj}\right) + Var\left(\overline{Y}_{b0}\right)\right]}}$$
$$= \frac{\frac{1}{n_{ej}n_{bj}}\sum^{n_{ej}}\sum^{n_{bj}} Cov\left(Y_{ej}, Y_{bj}\right) + \frac{1}{n_{e0}n_{b0}}\sum^{n_{e0}}\sum^{n_{b0}} Cov\left(Y_{e0}, Y_{b0}\right)}{\sqrt{\frac{\sigma_e^2}{n_{ej}} + \frac{\sigma_e^2}{n_{e0}}}\sqrt{\frac{\sigma_b^2}{n_{bj}} + \frac{\sigma_b^2}{n_{b0}}}}$$
$$= \frac{\left(\frac{n_j}{n_j^2} + \frac{n_0}{n_0^2}\right)\rho\sigma_e\sigma_b}{\sqrt{\frac{\sigma_e^2}{n_j} + \frac{\sigma_e^2}{n_0}}\cdot\sqrt{\frac{\sigma_b^2}{n_j} + \frac{\sigma_b^2}{n_0}}}$$
$$= \frac{\left(\frac{1}{n_j} + \frac{1}{n_0}\right)\rho}{\sqrt{\frac{1}{n_j} + \frac{1}{n_0}}\cdot\sqrt{\frac{1}{n_j} + \frac{1}{n_0}}} = \rho,$$

Note that this correlation coefficient is independent of the sample sizes.

Result 3: Given different doses j and p, where $j, p = 1, 2, \cdots, m, j \neq p$, The correlation coefficient between test statistics of efficacy endpoint and biomarker

$$\rho_{eb2} = Corr(E_j, B_p)$$
$$= Corr\left(\overline{Y}_{ej} - \overline{Y}_{e0},\ \overline{Y}_{bp} - \overline{Y}_{b0}\right)$$
$$= \frac{Cov\left(\overline{Y}_{ej} - \overline{Y}_{e0},\ \overline{Y}_{bp} - \overline{Y}_{b0}\right)}{\sqrt{Var\left(\overline{Y}_{ej} - \overline{Y}_{e0}\right)Var\left(\overline{Y}_{bp} - \overline{Y}_{b0}\right)}}$$
$$= \frac{Cov\left(\overline{Y}_{e0}, \overline{Y}_{b0}\right)}{\sqrt{\left[Var\left(\overline{Y}_{ej}\right) + Var\left(\overline{Y}_{e0}\right)\right]\cdot\left[Var\left(\overline{Y}_{bp}\right) + Var\left(\overline{Y}_{b0}\right)\right]}}$$
$$= \frac{\frac{n_0}{n_0^2}\rho\sigma_e\sigma_b}{\sigma_e\sigma_b\sqrt{\frac{1}{n_j} + \frac{1}{n_0}}\cdot\sqrt{\frac{1}{n_p} + \frac{1}{n_0}}}$$
$$= \frac{\frac{1}{n_0}\rho}{\sqrt{\frac{1}{n_j} + \frac{1}{n_0}}\cdot\sqrt{\frac{1}{n_p} + \frac{1}{n_0}}} = \sqrt{\frac{n_j n_p}{(n_0 + n_j)(n_0 + n_p)}}\rho.$$

For balanced case, i.e., all groups have the same sample size $n_0 = n_1 = \cdots = n_m = n$, the correlation coefficient of ρ_{eb2} reduces to

$$\rho_{eb2} = \frac{1}{2}\rho.$$

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